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324712 (24)

BE (7th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical

ELECTRICAL MACHINES - III

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt part a and any two from parts b, c and d from each question. Part a is of two marks and parts b, c and d of seven marks each.

Q. 1. (a) Why d axis reactances are only involved during three phase short circuit any an unloaded alternator.

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(b) In a three phase synchronous machine show that three phase stator windings with variable inductances can be converted to two windings with constant inductances.

(c) A three phase short circuit occurs at terminals of an unloaded alternator. Discuss the nature of current flowing in armature and field with the explaination of different reactances controlling currents. Q. 2.

(d) A two pole Y connected alternator has uniform air gap. The stator winding has synchronous inductance of 0.01 henry per phase and $r_a = 0$. The rotor winding inductance is 20 henrys and resistance of 10 ohms. It is connected with dc voltage

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source of 100 V. The maximum mutual inductance between stator and rotor is 2 henry. The machine is supplying power to a three phase load at unity p.f. and 500 radians per second angular frequency at phase voltage (terminal) of 1000 V. What is electrical power output in kW.

Q. 2. (a) State the condition of power invariance during transformation.

(b) For a three phase two pole induction motor show that d, q flux linkages for stator and rotor can be written

$$\lambda_{1d} = L_{11} i_{1d} + L_{12} i_{2d} \qquad \lambda_{1q} = L_{11} i_{1q} + L_{12} i_{2q}$$

$$\lambda_{2d} = L_{22} i_{2d} + L_{12} i_{1d} \qquad \lambda_{2q} = L_{22} i_{2q} + L_{12} i_{1q}$$

where notations have their usual meanings.

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(c) Show that developed torque of three phase p pole induction motor can be written as :

$$T = \frac{3}{2} \cdot \frac{P}{2} \left(\lambda_{2d} i_{2q} - \lambda_{2q} i_{2d} \right) \text{Nm.}$$

(d) The parameters of three phase induction motor are :

$r_1 = 0.7\ \Omega$, $x_1 = 1.3\ \Omega$, $r_2' = 0.4\ \Omega$, $x_2' = 0.6\ \Omega$

Neglect magnetising branch. If supply voltages are $V_a = 200\angle 0$, $V_b = 250\angle -100°$, $V_c = 300\angle 100°$, find developed torque in synchronous watts at slip of 5%.

Q. 3. (a) What is drag cup motor.

(b) Develop the equivalent circuit of single phase induction motor from double revolving field theory

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(c) A four pole 50 Hz single phase induction motor has $r_2' = 0.4\ \Omega$, $X_2' = 4\ \Omega$. If the motor is running at 1200 r.p.m. determine the frequencies of rotor current components and relative magnitudes of forward and backward fluxes and torques. Neglect magnetising current and stator impedances.

(d) A two phase a.c. servomotor supplied from balanced two phase supply voltages of 1 pu develop torque of 1 pu at slip = 0.5 and torque of 1.5 pu at slip = 1.5 p u The motor has been running at steady slip of 0.5 with 1 pu balance voltages when voltage to control phase is suddenly reduced to zero. Calculate the torque develop by servomotor.

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Q. 4. (a) Why universal motors are run at very high speeds.

(d) Draw the phasor diagram of wound rotor 3ϕ induction motor with injected emf E_J in rotor at angle β. Show that power transferred to secondary = secondary I^2R loss + power from rotor to auxiliary machine + mechanical power output.

(c) Explain the principler of commutator frequency converter and show that frequency of brush f_b is given by

$$f_b = \left(1 + \frac{N_r}{N_s}\right) f$$

where N_r = rotor speed, f = supply frequency, N_s is synchronous speed.

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(d) A two pole 0.1 HP, 50 Hz, 230 V series motor having salient poles has following constants: Total R = 23.6 Ω, Total leakage reactance = 68 Ω Mutual reactances in d and q axes 121 Ω and 72 ohms respectively. Friction and windage loss = 40 W. Calculate current power factor and output when running at 6000 rpm.

Q. 5. (a) Write the classification of synchros.

(b) Describe the construction and working principles of reluctance motor. Show that it can develop torque at only one speed.

(c) Describe the construction of three pulse brushless dc motor. Show that output torque

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of motor is independent of rotor angular position if phase current vary sinusoidally.

(d) A three stack VRSM has 16 teeth on both rotor and stator. Find angular displacement between stator stack for achieving rotation of rotor at speed of 810 rpm. Find frequency of currents given to stator phase winding.

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